

SYREEN CHAIN

Technical Whitepaper

A high-performance Layer 1 blockchain built for decentralized finance with on-chain trading, deflationary tokenomics, and cross-chain connectivity.

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<https://syreen.com>

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1. Abstract

Syreen Chain is a purpose-built Layer 1 blockchain designed for decentralized finance (DeFi) applications. Built on Cosmos SDK v0.53.5 and CometBFT consensus, Syreen delivers ~375ms block times, an integrated AMM decentralized exchange with order book trading, and a deflationary token model that permanently burns 80% of all transaction fees.

The chain features full EVM compatibility (Chain ID 79733), enabling Solidity smart contract deployment without modification. IBC protocol support connects Syreen to the broader Cosmos ecosystem for seamless cross-chain asset transfers.

With a genesis supply of **100,000,000 SYR** tokens and adaptive inflation between 0.5–2%, Syreen becomes net-deflationary as network usage grows. The protocol includes a hybrid AMM and order book DEX with limit orders, stop-loss, take-profit, smart order routing, concentrated liquidity, referral rewards, and copy trading — all implemented as native protocol modules.

2. Introduction

2.1 The Problem

Existing blockchain platforms force DeFi applications to choose between performance, security, and interoperability. Ethereum offers a mature ecosystem but suffers from high fees and 12-second block times. Solana provides speed but lacks native cross-chain communication. Cosmos chains offer IBC interoperability but often lack integrated DeFi primitives.

Most blockchain fee models are purely inflationary — transaction fees go entirely to validators with no mechanism to offset token supply growth. Smart contract-based DEXes introduce additional risk vectors: upgradeable proxy contracts, admin key compromises, and gas-intensive operations.

2.2 The Solution

Syreen Chain embeds core DeFi infrastructure directly into the protocol layer:

- **Integrated DEX** — AMM + order book as a native module with lower gas, atomic composability, and protocol-level security.
- **Intent-Based Trading** — Users express trading goals as on-chain intents. The protocol finds optimal execution paths via solver competition.
- **Deflationary Fee Model** — 80% of every transaction fee is permanently burned.
- **Full EVM + IBC** — Deploy Solidity contracts unchanged while accessing cross-chain liquidity.
- **MEV Protection** — Protocol-level commit-reveal ordering prevents frontrunning and sandwich attacks.

3. Architecture

3.1 Consensus Layer

Syreen uses CometBFT (formerly Tendermint) as its consensus engine:

- **Instant Finality** — Blocks are final once committed. No probabilistic finality.
- **~375ms Block Time** — Sub-second block production (~84M blocks per year).
- **BFT Tolerance** — Tolerates up to 1/3 of validators being Byzantine.
- **Delegated Proof of Stake** — Up to 100 active validators.

3.2 Application Layer

Built on Cosmos SDK v0.53.5 with a modular architecture where each protocol feature is an independent module with its own state, message handlers, and query endpoints.

3.3 Module Architecture

Module	Purpose	Status
dex	AMM + order book decentralized exchange	Active
clmm	Concentrated liquidity market maker	Active
intent	Intent-based transaction execution	Active
feemarket	EIP-1559 dynamic fees with 80% burn	Active
evm	Ethereum Virtual Machine compatibility	Active
tokenfactory	Permissionless token creation	Active
mevprotection	Transaction ordering fairness	Active
ibc	Inter-Blockchain Communication	Active
abstractaccount	Account abstraction & recovery	Active
compute	Smart contract engine (WASM)	Active
identity	On-chain identity	Active
payments	Payment processing	Active
vault	Auto-compounding vaults	Active
portfolio	Portfolio tracking & competitions	Active

perps	Perpetual futures	Regulatory Hold
options	Options trading	Regulatory Hold
lending	Lending & borrowing	Regulatory Hold
farming	Liquidity incentives	Regulatory Hold
launchpad	Token launch platform	Regulatory Hold
flashloan	Flash loans	Regulatory Hold
predict	Prediction markets	Regulatory Hold
aiagent	AI trading agents	Regulatory Hold

Modules marked "Regulatory Hold" are built and included in the binary but disabled via on-chain governance toggles. They can be activated through governance proposals when regulatory clarity is achieved.

4. Decentralized Exchange (DEX)

4.1 Automated Market Maker

The Syreen DEX implements a constant-product AMM ($x*y=k$) at the protocol level, providing lower gas consumption, atomic composability, and protocol-level security compared to smart contract DEXes.

4.2 Swap Mechanics

$$\text{Output Amount} = R_b \times \Delta a / (R_a + \Delta a)$$

Default swap fee: **0.3%**. Of this: 66.7% to liquidity providers, 33.3% to protocol (subject to 80% burn).

4.3 Liquidity Pools

Anyone can create a pool by providing equal value of two tokens. Minimum initial liquidity: 1,000,000 base units. LP tokens represent pool share and can be redeemed at any time.

4.4 Order Book

Full order book with: Limit Orders, Market Orders, Stop-Loss, Take-Profit, Stop-Loss Limit, Take-Profit Limit. Time-in-force: GTC, IOC, FOK. Orders matched in BeginBlock before AMM operations.

4.5 Smart Order Routing

- **Multi-hop routing** — Up to 3 pool hops for best price path
- **Order splitting** — Split across up to 5 routes to minimize price impact
- **Price impact scoring** — Routes scored by expected output, fees, and impact

4.6 Concentrated Liquidity (CLMM)

Uniswap v3-style capital-efficient liquidity positions. Providers concentrate capital within specific price ranges for dramatically improved efficiency.

4.7 Referral System

Tier	Referrals	Referrer Share	Trader Discount
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Bronze	0+	5%	5%
Silver	10+	7%	5%
Gold	50+	10%	7%
Platinum	200+	12%	10%

4.8 Volume-Based Fee Tiers

Tier	Min 30-Day Volume	Fee Discount
Standard	0 SYR	0%
Silver	50,000 SYR	10%
Gold	250,000 SYR	20%
Platinum	1,000,000 SYR	30%
Diamond	5,000,000 SYR	40%

4.9 Copy Trading

Automatically replicate trades of successful traders. Features: budget limits per trade, total budget caps, configurable copy ratios, automatic PnL tracking, auto-unfollow on poor performance.

5. Intent System

Users express complex trading goals as on-chain intents rather than individual transactions. An intent specifies what the user wants to achieve without specifying the exact execution path.

Parameter	Value
Minimum solver stake	1,000 SYR
Solving window	10 blocks (~4 seconds)
Max solutions per intent	20
Solver slash fraction	10%
Intent expiry	20,000 blocks (~2.1 hours)

Solvers stake SYR and compete to find optimal execution paths. The protocol selects the best solution and rewards the winning solver. Invalid solutions result in 10% stake slashing.

6. Fee Market & Burn Mechanism

6.1 EIP-1559 Fee Market

Parameter	Value
Default base fee	0.01 usyreen/gas
Min / Max base fee	0.0001 / 1,000 usyreen/gas
Change denominator	8 (max 12.5%/block)
Elasticity multiplier	2

6.2 Fee Lanes

Lane	Multiplier	Max Block Gas	Purpose
Default	1.0x	50M	Standard transactions
DeFi	1.5x	30M	DEX swaps, liquidity
IBC	1.2x	20M	Cross-chain transfers
Governance	0.8x	10M	Voting (discounted)

6.3 The 80% Burn

```
burn_amount = total_fees × 0.80 | validator_reward = total_fees × 0.20
```

Burned tokens are permanently removed from supply via the bank module's BurnCoins function.

6.4 Deflationary Analysis

At 1% inflation on 100M supply, ~1M SYR minted per year. Burn rate depends on activity:

Network TPS	Annual Burn	Net Supply Change
10 TPS	126K SYR	+874K (inflationary)
~50 TPS	~1M SYR	~0 (net zero)
100 TPS	2.5M SYR	-1.5M (deflationary)
500 TPS	12.6M SYR	-11.6M (deflationary)

Net-zero threshold: **~50 TPS**. Beyond this, SYR becomes deflationary.

7. EVM Compatibility

Parameter	Value
EVM Chain ID	79733
RPC Endpoint	JSON-RPC 2.0 compatible
Supported EIPs	EIP-1559, EIP-155, EIP-2930
Gas Token	SYR (usyreen)
EVM Version	Paris (pre-Shanghai)

Deploy Solidity contracts with Hardhat, Foundry, Remix, or MetaMask. Contracts must target EVM version Paris (no PUSH0). EVM contracts interact with native modules through precompiled contracts.

8. Cross-Chain Communication (IBC)

Syreen implements IBC-Go v10, enabling trustless cross-chain asset transfers across the Cosmos ecosystem. Assets received via IBC are represented as IBC denominations and can be traded on the Syreen DEX.

9. Token Factory

Create new tokens without writing smart contracts. Tokens follow `factory/{creator}/{subdenom}` format. Creators have admin authority: mint, burn, and transfer admin rights.

10. Account Abstraction

- **Smart Accounts** — Multi-owner with configurable threshold signatures
- **Session Keys** — Temporary keys with scoped permissions (max 24 hours)
- **Social Recovery** — Up to 7 guardians, 48-hour delay period
- **Gas Sponsorship** — Sponsors pay gas for other users (gasless onboarding)
- **Batch Execution** — Multiple transactions atomically in one message

11. Tokenomics

11.1 Genesis Distribution

Total supply: **100,000,000 SYR**

Allocation	%	Amount	Purpose
Liquidity Pools	30%	30M SYR	DEX pool seeding, deep order books
Trading Rewards	20%	20M SYR	Incentivize active traders, fee rebates
Founder / Team	15%	15M SYR	Vested (see 11.2)
Treasury	10%	10M SYR	Protocol development, operations
Marketing / Listings	10%	10M SYR	CEX listings, partnerships
Community / Airdrops	9.6%	9.6M SYR	User acquisition, rewards
Staking Reserve	5%	5M SYR	Future validator onboarding
Genesis Validators	0.4%	400K SYR	Initial validator stakes (4 × 100K)

11.2 Founder Vesting Schedule

Enforced on-chain via `PeriodicVestingAccount` — the blockchain prevents spending unvested tokens.

Period	Duration	Unlocked
Cliff	Months 0–6	0%
First unlock	Month 7	~4.2%
Linear vesting	Months 8–30	+4.2%/month
Fully vested	Month 30	100%

Locked tokens **can still be delegated** for staking rewards. Rewards are immediately spendable.

11.3 Inflation & Staking

Adaptive inflation: 0.5% min to 2% max, targeting 67% bonded ratio. The inflation rate self-adjusts — rising when staking drops below 67% to incentivize delegation, and falling when it exceeds 67% to encourage liquidity. Rate change: 0.25% per year. Validators earn block rewards (inflation) plus 20% of transaction fees. Minimum commission: 5%. Unbonding: 21 days.

12. Governance

Parameter	Value
Minimum deposit	100 SYR
Voting period	10 minutes
Quorum	33.4%
Pass threshold	50%
Veto threshold	33.4%

Proposals can modify chain parameters, upgrade modules, allocate community funds, or make protocol changes. Votes weighted by staked SYR.

13. Security

Violation	Penalty	Jail Duration
Double signing	5% slashed	Permanent
Downtime (<50% signed)	1% slashed	10 minutes

MEV protection via commit-reveal scheme (3-block commit, 1-block reveal) prevents frontrunning and sandwich attacks at the protocol level.

14. Network Parameters

Parameter	Value
Chain ID (Cosmos)	syreen-1
Chain ID (EVM)	79733
Block time	~375ms
Max validators	100
Bond denom	usyreen (1 SYR = 1,000,000 usyreen)
Unbonding period	21 days
Fee burn ratio	80%
Inflation range	0.5% – 2%
Goal bonded	67%
Genesis supply	100,000,000 SYR
Blocks per year	~84,153,600

15. Roadmap

Completed

- Live blockchain with ~375ms blocks and 4 validators
- On-chain DEX with AMM pools, order book, stop-loss, take-profit

- Smart order routing with multi-hop and order splitting
- 80% fee burn mechanism active
- EVM compatibility (Chain ID 79733) + WASM smart contracts
- IBC module, Token Factory, Web wallet, Block explorer
- Founder vesting enforced on-chain via PeriodicVestingAccount
- 100M SYR genesis with trading platform tokenomics
- Full SDK v0.53.5 + IBC-Go v10 + self-balancing inflation (0.5%–2%)

Next: Exchange & Listings

- IBC bridge to Osmosis mainnet
- SYR listed on Osmosis DEX
- CoinGecko and CoinMarketCap listings
- Validator recruitment and community building

Upcoming: Trading Platform

- Professional trading platform with advanced charting
- Keplr wallet integration
- Cross-chain asset trading via IBC

Future: Ecosystem Growth

- CEX listings, 50+ validators, mobile trading app
- Cross-chain bridges for ETH and BTC assets
- Developer grants program
- Regulatory approval for disabled modules (perps, options, lending)

16. Conclusion

Syreen Chain represents a new approach to blockchain-based trading: purpose-built infrastructure where every protocol decision is optimized for DeFi. By embedding the DEX, intent system, fee burning, and MEV protection directly into the chain's state machine, Syreen eliminates the complexity, gas overhead, and security risks of smart contract-based alternatives.

With 100M SYR genesis supply, on-chain founder vesting, 80% fee burn, self-balancing inflation, and ~375ms block times, Syreen is positioned as a high-performance trading platform built for the next generation of decentralized finance.